

Medication Reconciliation Tool

CMT: CMT-0018 (SR)

MCG Health
Shared Resources
28th Edition

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Medication Reconciliation

Introduction

Medications are a significant part of the condition treatment regimen. As patients move through the healthcare continuum, medication errors can be made, putting patient safety at risk.⁽¹⁾⁽²⁾ Taking a medication history at each point of the healthcare continuum can ensure that an accurate medication list is constructed and communicated to all people involved in the patient's care.

The goal of medication reconciliation is to prevent adverse drug events (ADEs) by identifying and correcting medication discrepancies.^[A] A discrepancy is the identification of an unintended difference between two medication regimens (eg, the patient's home regimen vs new orders written on admission) in one of 4 components: name, dosage, route, or frequency.⁽³⁾

Medication reconciliation is a process of identifying the most accurate list of all medications a patient is taking, including medication name, purpose, dosage, frequency, route, prescribing physician, and any over-the-counter (OTC) medications.⁽⁴⁾ It involves comparing the patient's current list of medications against the physician's admission, transfer, or discharge orders and in the outpatient setting (eg, the specialist's office, home care, community pharmacies, or emergency department). The reconciliation should also include an assessment of each medication for correlation to a disease state or medical condition.⁽⁵⁾ If medication discrepancies are identified at any point during the systematic reviews, the prescribing physician and the primary care physician (PCP) should be notified so that an accurate medication order is identified and communicated.

National accreditation organizations such as The Joint Commission and the Institute for Healthcare Improvement (IHI) have initiated standards for medication reconciliation. The goal of these standards is to be a patient-centered function to improve medication safety.⁽⁶⁾⁽⁷⁾ Both organizations focused on ADEs occurring in the inpatient setting or as a reason for admission to the acute care setting.

Medication Reconciliation Process

- **Verification**⁽⁸⁾
 - Obtain medication lists from all treating providers.
 - Collect medication information from patient or caregiver; questions include⁽⁵⁾⁽⁹⁾:
 - Who are all the healthcare providers that order medications?
 - What are all the medications that are being taken?
 - What are the medications for?
 - What medications are taken every day?
 - What medications are taken as needed? What are the triggers for taking these medications?
 - What reminders are used to take medication (eg, timers, calendars)?
 - How long has medication been taken (eg, is it a newly prescribed medication, has patient taken this medication for years)?
 - How often are medication doses missed? If patient has voluntarily missed or stopped taking a medication:
 - Why were the medications missed or stopped?
 - Was the doctor notified? If not, why?
 - Are there any medications that are given by other people or in clinical settings (eg, with procedures such as dialysis, vitamin B12 injections)?
 - Are any over-the-counter (OTC) or herbal medications taken? If so, what are the OTC or herbal medications, and how often are they taken?
 - What pharmacies are used to have prescriptions filled?
 - Are there any allergies to medications?
 - What medications cause reactions?
 - What are the reactions?
 - How does the patient order refills?
 - Are refills called in by the doctor?
 - Are paper refill prescriptions dropped off at the pharmacy?
 - Are refills ordered through electronic service (eg, telephone, computer system)?
 - Will the pharmacy contact the doctor for renewals?
 - How are medications paid for?
 - How are medications obtained? Are they:
 - Picked up at the pharmacy?
 - Are there issues with picking up the medication (eg, patient depends on caregivers to pick them up, patient does not have transportation to pick them up)?
 - Delivered by the pharmacy?
 - Delivered by a mail order service?
 - Medications are appropriate for patient's disease state or medical condition.⁽¹⁰⁾
 - If patient has chronic condition, is there medication they should be taking that they are not (eg, hypoglycemics for diabetes)?
 - Is the patient taking medication that is not usual for their medical situation?
- **Comparison**
 - Review current medication available with the list of medications to be prescribed at next level of care or care transition.
- **Clarification**⁽⁸⁾
 - Ensure medications and doses are appropriate.
 - Review and address risk factors for adverse drug events (ADEs); examples include⁽⁵⁾⁽¹¹⁾:
 - Advanced age
 - Polypharmacy
 - Chronic condition
 - Patient's place of residence
 - Social support
 - Therapeutic duplication, drug-drug interaction, brand-generic duplication, prescription drug to which the patient is known to be allergic
- **Reconciliation**
 - Document changes in the orders at all encounter points; encounter points include⁽⁴⁾:
 - Specialist visit
 - Patient's list with primary care provider's (PCP) list
 - Patient's list with specialist's medical record
 - Home healthcare
 - Patient's list with discharge orders
 - Discharge orders with medications found in the home
 - Patient's medication list in PCP's records
 - Patient's current community pharmacy profile
 - Emergency department
 - Patient's list with patient and caregiver interviews

- Patient's list with PCP's list
- Facility admission (eg, acute care, recovery facility, inpatient rehabilitation facility)(1)
 - Patient's list with initial primary healthcare orders
 - PCP's list if elective or direct admission
- Unit transfer
 - Current list with transfer orders on new unit
- Facility discharge
 - Facility medication list with discharge orders
- Facility transfer (eg, needed specialty services or additional acute care needs)
 - Prior facility discharge medication list with current facility admission medication list
- Outpatient visit after discharge
 - Medications currently being taken by patient (patient should bring all medication containers to post-hospital visit)
 - Facility discharge medication orders(1)
 - Patient's medication list in PCP records
 - Patient's current community pharmacy profile
- Preoperative screen(12)
- Report discrepancies to ordering physician and primary care physician.(3)
- **Communication(4)**
 - Provide new reconciled list to all appropriate providers, healthcare personnel, and caregivers. Healthcare personnel includes:
 - PCP
 - Specialist physicians
 - Dentists
 - Home care personnel
 - Community pharmacy
 - Therapists
 - Confirm that reconciled list sent to next provider of service.
 - Confirm patient or caregiver can acquire all of patient's medications.
 - Educate patient and caregiver on medication information.

Patient and Caregiver Education

- Provide written and verbal instruction to patient and caregiver on medication details, including(4)(5):
 - Names of medication, including generic and brand names
 - Purpose of each medication
 - Dosage, including changes as appropriate
 - How to administer or take drug
 - Route (eg, oral)
 - With or without foods or liquids (eg, milk)
 - Schedule (taking into account mealtimes, ADL)
 - Expected effects if drug is working and expected time for effects to occur
 - Potential side effects
 - Anticipated length of therapy
 - Appropriate action if side effects or allergic reaction occurs(8)
 - When to stop medication
 - When to call prescribing provider (eg, primary care physician (PCP), surgeon, nurse practitioner)
 - When to go to the emergency department
 - Consequences of improper use or discontinuation without physician approval
 - Importance of refilling prescription before medications run out and how and where to refill or order(13)
 - Storage requirements (eg, refrigeration)
 - Interactions with over-the counter (OTC) medications (eg, cold remedies), herbal preparations, food or vitamin supplements
 - Proper disposal of unused or expired prescription and OTC medication(13)
- Emphasize importance of presenting personal medication list to **all** providers at each care transition (eg, all specialists, healthcare providers).(4)
- Techniques for education and reinforcement include:
 - Teach back.(14) See Teach Back Tool [🔗](#) SR.
 - Group medications on list by purpose rather than in alphabetical order (eg, heart or blood pressure medications).
 - Use illustrations to promote understanding.(15)
 - Label or color-code medication bottles.
 - Pill cards with a picture of the pill next to the medication detail (ie, name, purpose, dosage, when to take)
 - Time of day for taking medication (eg, analog clock with time, sunup or sundown, bedtime)
 - Review medication list during postdischarge phone call. See Post-Acute Care Follow-Up Call Tool [🔗](#) SR.
 - See Education: Patient, Family, and Caregiver [🔗](#) SR.

Medication Reconciliation Worksheet

PATIENT NAME:

DATE:

TRANSITIONING TO (circle one):

Emergency department

Acute care

Observation

Unit transfer

LTACH

Inpatient rehabilitation facility (IRF)

Recovery facility

Home healthcare service

Home

New residence (eg, assisted living)

Other residence (eg, family member's home)

Provider office

Outpatient clinic

Ambulatory surgery

MEDICATIONS

DISCREPANCY RESOLUTION

Include prescribed medications, OTC, and herbal preparations.

Current Order					Consistent with transition list? (Yes/No) If "No" complete Discrepancy Resolution	Physicians notified Name/date	Reconciled Order					Accurate list transmitted to all providers (check when complete)	Accurate written list given to patient with verbal instructions (check when complete)
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Drug Name	Dose	Route	Frequency	Prescriber		Drug Name	Dose	Route	Frequency	Prescriber			
1.													
2.													
3.													
4.													
5.													
6.													
7.													
8.													
9.													
10.													

Resources

- Additional information, resources, and tools include[C]:
 - After Hospital Care Plan from the Re-Engineered Discharge (RED) program, Boston Medical Center at www.bu.edu/fammed/projectred(16)
 - Medication Reconciliation Elements Form from the National Transitions of Care Coalition at www.ntocc.org(17)
 - Improving transitions "How-to Guides" from the Institute for Healthcare Improvement at www.ihl.org(18)(19)(20)(21)

1. Fischer SH, et al. Standardized assessment of medication reconciliation in post-acute care. *Journal of the American Geriatrics Society* 2022;70(4):1047-1056. DOI: 10.1111/jgs.17655. [Context Link 1, 2, 3] View abstract...

2. Dautzenberg L, et al. Medication review interventions to reduce hospital readmissions in older people. *Journal of the American Geriatrics Society* 2021;69(6):1646-1658. DOI: 10.1111/jgs.17041. [Context Link 1] View abstract...

3. Adverse drug reactions and medication errors. In: Burchum JR, Rosenthal LD, editors. *Lehne's Pharmacology for Nursing Care*. 11th ed. Elsevier; 2022:63-73. [Context Link 1, 2, 3]

4. Gleason K, Brake H, Agramonte V, Perfetti C. Medications at Transitions and Clinical Handoffs (MATCH) Toolkit for Medication Reconciliation. (Prepared by the Island Peer Review Organization, Inc., under Contract No. HHS2902009000 13C.) AHRQ Publication No. 11(12)-0059 [Internet] Agency for Healthcare Research and Quality. 2012 Aug Accessed at: <https://www.ahrq.gov/>. [created 2011; accessed 2023 Oct 09] [Context Link 1, 2, 3, 4, 5]

5. Smith DM, Friend C, Reeve J. Polypharmacy and rationalisation of medications. *InnovAIT* 2020;13(2):87-93. [Context Link 1, 2, 3, 4]

6. National Patient Safety Goals. 2024 National Patient Safety Goals [Internet] Joint Commission on Accreditation of Healthcare Organizations. Accessed at: https://www.jointcommission.org/standards_information/npsgs.aspx. Updated 2024 [accessed 2023 Nov 28] [Context Link 1]

7. 5 Million Lives Campaign. [Internet] Institute for Healthcare Improvement. Accessed at: <https://www.ihl.org/Engage/Initiatives/Completed/5MillionLivesCampaign/Pages/default.aspx>. Updated 2023 [accessed 2023 Aug 15] [Context Link 1]

8. Ostendorf WR. Preparation for safe medication administration. In: Perry AG, Potter PA, Ostendorf WR, editors. *Nursing Interventions and Clinical Skills*. 7th ed. Elsevier; 2020:551-567. [Context Link 1, 2, 3]

9. Managing Medicines for Adults Receiving Social Care in the Community. NICE Guidance NG67 [Internet] National Institute for Health and Care Excellence. 2017 Mar Accessed at: <https://www.nice.org.uk/guidance/>. [accessed 2023 Aug 08] [Context Link 1]

10. Sochacki S. Adherence. In: Larsen PD, editor. *Lubkin's Chronic Illness: Impact and Intervention*. 11th ed. Jones & Bartlett Learning; 2022:269-306. [Context Link 1]

11. Nursing care of the elderly client. In: Haugen N, Galura SJ, editors. *Ulrich & Canale's Nursing Care Planning Guides*. 8th ed. St. Louis, MO: Elsevier; 2020:806-844. [Context Link 1]

12. Neil JA. Preoperative care. In: Harding MM, Kwong J, Hagler D, Reinisch C, editors. *Lewis's Medical-Surgical Nursing: Assessment and Management of Clinical Problems*. 12th ed. St. Louis, MO: Mosby; 2023:335-349. [Context Link 1]

13. The nursing process in drug therapy and patient safety. In: Tucker RG, editor. *Karch's Focus on Nursing Pharmacology*. 9th ed. Wolters Kluwer; 2023:47-57. [Context Link 1, 2]

14. Medication adherence. In: Perez R, Rogers S, Prince M, Fraser K, editors. *Case Management Adherence Guidelines*. 2020 ed. Case Management Society of America; 2022:57-74. [Context Link 1, 2]

15. Martinez-Kratz M. Guide to planning care. In: Makic MB, Martinez-Kratz MR, editors. *Ackley and Ladwig's Nursing Diagnosis Handbook*. 13th ed. Elsevier; 2023:125-1051. [Context Link 1]

16. The Project RED Toolkit. [Internet] Boston University School of Medicine. Accessed at: <https://www.bu.edu/fammed/projectred/toolkit.html>. Updated 2014 [accessed 2023 Oct 10] [Context Link 1]

17. Medication Reconciliation Elements. [Internet] National Transitions of Care Coalition. 2008 Apr Accessed at: <https://www.ntocc.org>. [accessed 2023 Aug 10] [Context Link 1]

18. Herndon L, Bones C, Bradke P, Rutherford P. How-to Guide: Improving Transitions from the Hospital to Skilled Nursing Facilities to Reduce Avoidable Rehospitalizations. [Internet] Institute for Healthcare Improvement. 2013 Jun Accessed at: <https://www.ihl.org/>. [accessed 2023 Oct 26] [Context Link 1]

19. Rutherford P, Nielsen GA, Taylor J, Bradke P, Coleman E. How-to Guide: Improving Transitions from the Hospital to Community Settings to Reduce Avoidable Rehospitalizations. [Internet] Institute for Healthcare Improvement. 2013 Jun Accessed at: <https://www.ihl.org/>. [accessed 2023 Oct 26] [Context Link 1]

20. Schall M, Coleman E, Rutherford P, Taylor J. How-to Guide: Improving Transitions from the Hospital to the Clinical Office Practice to Reduce Avoidable Rehospitalizations. [Internet] Institute for Healthcare Improvement. 2013 Jun Accessed at: <https://www.ihl.org/>. [accessed 2023 Oct 26] [Context Link 1]

21. Sevin C, Evdokimoff M, Sobolewski S, Taylor J, Rutherford P, Coleman EA. How-to Guide: Improving Transitions from the Hospital to Home Health Care to Reduce Avoidable Rehospitalizations. [Internet] Institute for Healthcare Improvement. 2013 Jun Accessed at: <https://www.ihl.org/>. [accessed 2023 Oct 26] [Context Link 1]

Footnotes

[A] Adverse drug event is defined as an injury resulting from a medication, including harm from the use of the medication. It can be in the form of an allergic reaction, unintended overdose, adverse effect, or vaccine reaction.(3) [A in Context Link 1]

[B] Education of patients and caregivers via the "teach back" method can reduce adverse drug events. Teach back is considered an effective technique for improving patient safety along with allowing ample time for questions.(14) [B in Context Link 1]

[C] There are many private and publically available resources and tools available to support medication reconciliation processes. Copyright or other permissions may be necessary prior to the use of these forms. Users should contact the sponsoring organization to obtain permission, if necessary. [C in Context Link 1]